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Euclidean Geometry In Mathematical Olympiads

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Russian Olympiad 1996 Problem 1.41



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#1

Point E and F are on side BC of convex quadrilateral $ABCD$ (with E closer than F to B). It is known that $\angle BAE = \angle CDF$ and $\angle EAF = \angle FDE$. Prove that $\angle FAC = \angle EDB$

My thoughts



I'm solving this by having 5 variables of angles, if you guys have more slick solution please share.

This post has been edited 1 time. Last edited by Vrangr, Jul 30, 2019, 8:08 AM

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#2

My friend, you are overcomplicating this very very simple problem. If you do not have this key observation you will not know what to do.

Hint



You only need to show that $ABCD$ is cyclic. Which quadrilateral is cyclic?

Hint 2



$AEFD$ is cyclic

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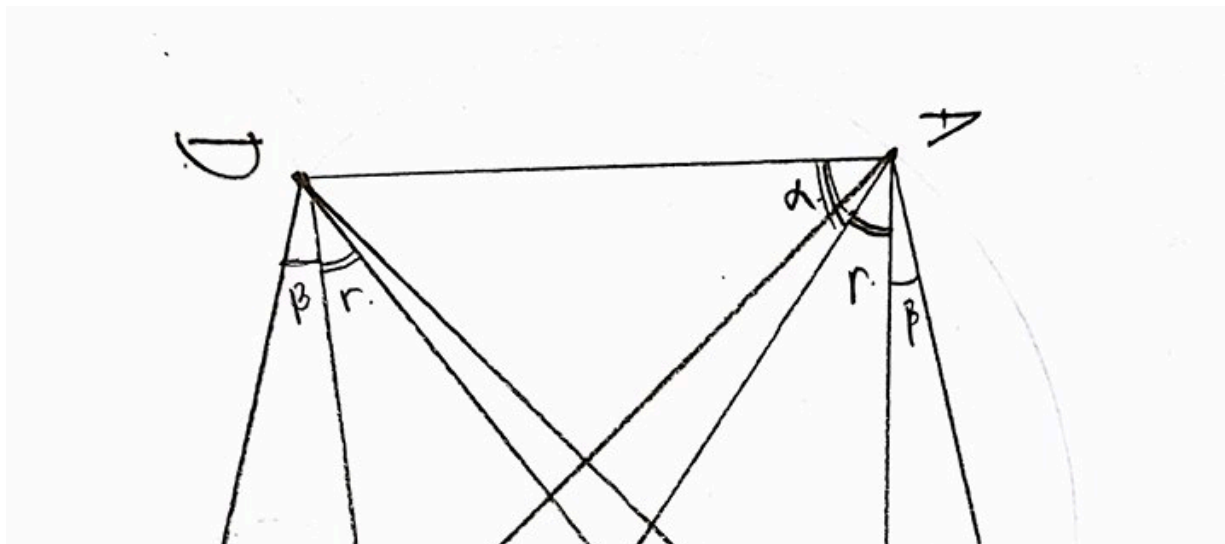
#3

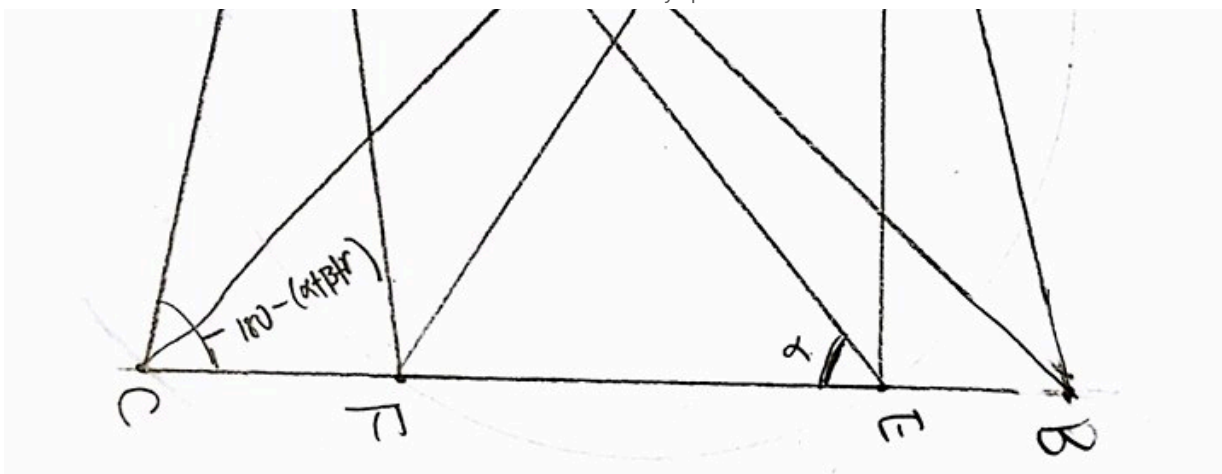
Solution



First, note $AEFD$ is cyclic. Let $\angle EAF = \angle FDE = \gamma$ and $\angle FDC = \angle BAE = \beta$. Denote $\angle DAF = \angle DEF = \alpha$. By the triangle sum for $\triangle DEC$, $\angle DCE = 180 - (\alpha + \beta + \gamma)$. Observe, $\angle DAB = \alpha + \beta + \gamma$. Hence, $\angle DAB + \angle DCE = 180$ and $ABCD$ is cyclic. Thus, $\angle CDB = \angle CAB$, and the result follows. ■

Attachments:





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#4

I definitely saw the cyclic quadrilateral in the observation though, but thanks for providing the solution